

An Autonomous Agent that Derives ADC Linker Rules from the Literature, Designs the Molecules, and Flags When the Field Disagrees

Elia Savino¹, Joshua W. Sin^{2,3}, Morgan G. L. Reigner¹, Miquèl À. Pérez-Puigdomènech², Derk H. W. ten Klooster¹

¹Noël Research Group, Van 't Hoff Institute for Molecular Sciences, University of Amsterdam, The Netherlands

²Laboratory of Artificial Chemical Intelligence (LIAC), EPFL, Lausanne, Switzerland

³Process Chemistry & Catalysis, F. Hoffmann-La Roche AG, Basel, Switzerland

Abstract

We report an autonomously derived, falsifiable prediction for antibody–oligonucleotide conjugate (ARC) design: a protease-cleavable Val-Cit linker is a viable alternative to the rigid non-cleavable sulfo-SMCC standard, because the modern clinical ARC literature has already moved that way. An agent reached this independently—it reads 31 primary papers and derives a design rule per payload class with a retrieval-grounded language model, and a leave-one-paper-out test shows the ARC rule is contested: withhold the rigid-non-cleavable siRNA paper and the agent derives protease-cleavable Val-Cit from the clinical antibody-oligonucleotide literature (DYNE-101/251, AOC-1001), lowering its confidence to 0.33 because the grounded evidence genuinely splits. Each rule parameterises a REINVENT4 objective that generates the linkers, and a Val-Cit ARC linker generated under the derived cleavable rule is recognised by cathepsin B (predicted K_d 10 nM), as tightly as the clinical Val-Cit substrate (35 nM) and unlike the rigid ARC design (109 nM)—an in-silico feasibility proof for the prediction. By contrast the cytotoxin and ISAC rules are robust (they recover at 0.97 and 0.83 when a paper is withheld), so the agent’s confidence tracks the literature’s actual consensus. Every reported value is provenance-tagged and a gate blocks placeholders from results, so the prediction is a grounded, calibrated conclusion a lookup-table system could not produce.

1 Introduction

Antibody conjugates couple antibody selectivity to a potent cargo, and the linker sets their therapeutic index [1, 3]. The linker requirement inverts with payload class: cytotoxins favour cleavable release [2], ISACs demand plasma stability [6, 7], and antibody–oligonucleotide conjugates (ARCs) were reported to favour rigid non-cleavable linkers [5]. An autonomous scientist should derive such rules from the literature, use them to design molecules, and report how confident it is—including recognising where the literature genuinely conflicts. We show all three, and the last surfaces a real, unresolved tension in ARC linker design (Figure 1) that we sharpen into a testable prediction:

Hypothesis (agent-derived, falsifiable): for antibody–oligonucleotide conjugates, a protease-cleavable Val-Cit linker matches or outperforms the rigid non-cleavable sulfo-SMCC standard on protease-mediated payload release. We derive it from the literature and test its structural feasibility in silico.

2 Methods

2.1 Reasoning that drives generation

For each payload class an agent (Claude Opus, always on) queries a store of 31 papers, retrieves passages with provenance, and extracts conjugate exemplars that must each cite a retrieved passage. It derives a rule (cleavage, rigidity, plasma-stability) from the grounded exemplars—with no hardcoded fallback, so a failed call is an explicit abstain—and the rule compiles into an ADC goal profile (cleavage regime, rotatable-bond window, per-term weights) that parameterises the REINVENT4 LinkInvent objective. That objective then generates the linkers (staged learning, remote GPU); the class trigger warhead (Val-Cit-PABC, Val-Ala-PABC or the rigid sulfo-SMCC cap) is welded into every molecule (Figure 3).

One molecule, end to end: a retrieved quote grounds the rule that generates the linker that cathepsin B recognises

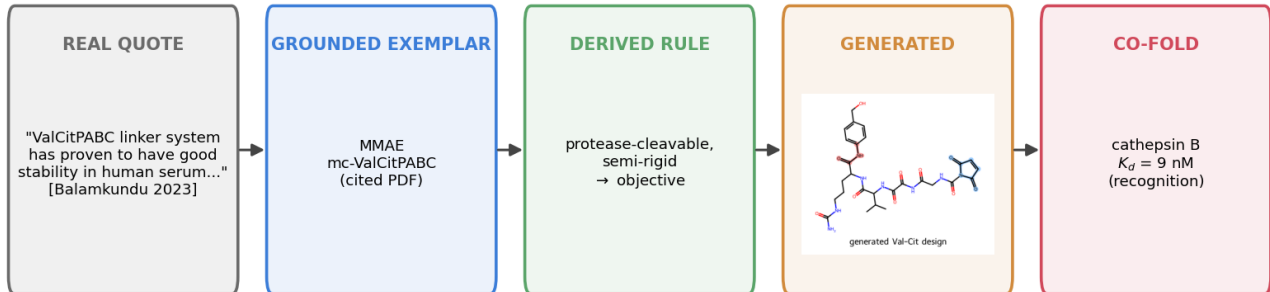


Figure 1: One molecule, end to end: a retrieved quote grounds the cytotoxin rule that generates a Val-Cit linker that cathepsin B recognises. Handle (blue), scissile bond (red) and spacer (green) are SMARTS-detected.

2.2 Confidence, scorers and provenance

Rule confidence is *disagreement-aware*: it is high only when the grounded exemplars are both plentiful and consistent, so a split field drives confidence down and is flagged contested. Synthesizability is the continuous Ertl SA_Score (the metric the generator optimises); the step-count survives only as a reported complexity index, not a route. Plasma stability is mechanism-resolved. Every value that reaches the manuscript is tagged *measured* (a real Boltz or REINVENT run), *llm*, or *heuristic*; a gate blocks any placeholder from a results claim (SI provenance table).

2.3 Held-out test and co-fold

We withhold a class’s key paper, re-derive the rule from the remaining corpus, record the prediction before revealing the withheld paper, and report agree/disagree with its evidence composition. We co-fold each designed linker with its real payload against cathepsin B (Boltz-2) as two co-present ligands—a recognition/plausibility check, not a covalent conjugate and not a claim of cleavage.

3 Results

3.1 The rules generate, and steer, the molecules

Each derived rule compiles into a distinct objective (Figure 3, left) that generates its linkers: the designs separate cleanly by class along the two rule knobs (Figure 3, right), ARC in a rigid non-cleavable region (4–6 rotatable bonds), cytotoxin and ISAC as cleavable flexible peptides (12–16). The molecules are made by the reasoning, not labelled after the fact.

3.2 Held-out test: robust rules and a contested one

The leave-one-paper-out test discriminates (Figure 2, Table 1). The cytotoxin and ISAC rules are robust: their grounded exemplars are unanimously cleavable, so withholding a paper leaves the derived rule and its high confidence intact. The ARC rule is contested. With the full corpus the top-retrieved siRNA papers give rigid non-cleavable. Withhold them, and from the clinical antibody-oligonucleotide literature (DYNE-101/251, AOC-1001) the agent derives protease-cleavable Val-Cit instead—its direction flips, and because the grounded evidence splits four cleavable to two non-cleavable, its disagreement-aware confidence falls to 0.33. Early cationic-assistance-free siRNA conjugates favour rigid non-cleavable linkers; clinical AOCs increasingly use cleavable Val-Cit. The agent surfaces that tension and lowers its confidence accordingly, rather than asserting a single rule.

3.3 An agent-derived hypothesis, tested in silico

The contested ARC result is a prediction, not just a caveat. From the clinical antibody-oligonucleotide literature the agent derives that ARC linkers can be protease-cleavable Val-Cit, against the rigid non-cleavable standard. We tested its structural feasibility: a Val-Cit ARC linker generated under the derived cleavable rule co-folds with cathepsin B at predicted K_d 10 nM (Figure 4), matching the clinical Val-Cit substrate (35 nM) and the cytotoxin Val-Cit design

Table 1: Leave-one-paper-out. Confidence is disagreement-aware; “evidence” is the grounded cleavable/non-cleavable split. The ARC rule flips direction and loses confidence because its evidence is genuinely divided.

Class	Withheld	Full-corpus	Held-out	Evidence	Outcome
Cytotoxin	Balamkundu 2023	cleavable (0.60)	cleavable (0.97)	8/0	recovers
Immunomodulator (ISAC)	imidazoquinoline ISAC	cleavable (0.83)	cleavable (0.83)	1/0	recovers
Oligonucleotide (ARC)	ARC siRNA 2025	non-cleavable (0.50)	cleavable (0.33)	4/2	flips, contested

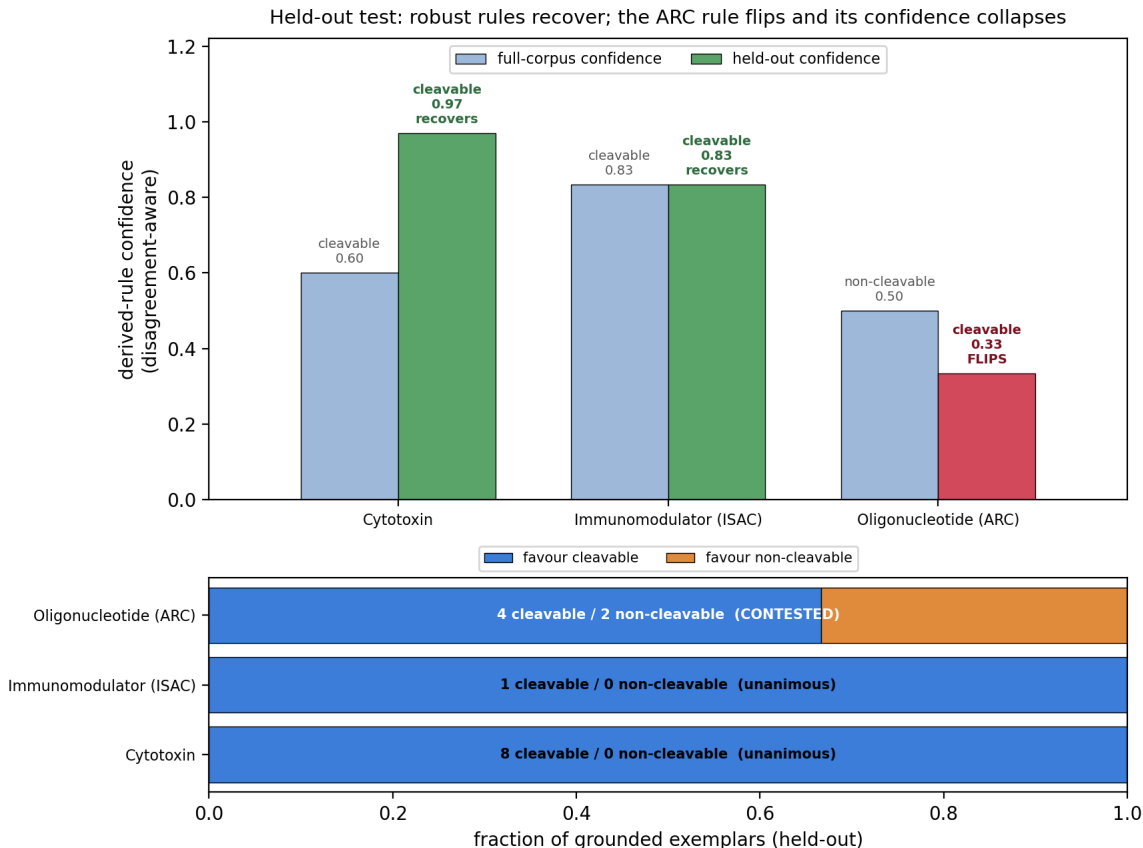


Figure 2: Held-out test. Top: full-corpus vs held-out confidence (disagreement-aware); the ARC rule flips (red) while cytotoxin/ISAC recover (green). Bottom: the grounded-evidence split that explains it—ARC is 4:2 (contested), the others unanimous.

(9 nM), while the rigid non-cleavable ARC design binds an order of magnitude more weakly (109 nM). The prediction is thus both literature-derived and structurally feasible, and directly falsifiable: a wet-lab comparison of Val-Cit versus sulfo-SMCC ARC linkers on protease-mediated release and potency would confirm or refute it.

3.4 Structural recognition

The rule-generated Val-Cit cytotoxin design co-folds tightly with cathepsin B (predicted K_d 9 nM), near the clinical mc-Val-Cit-PABC substrate (35 nM), while the rigid non-cleavable ARC design binds an order of magnitude more weakly (109 nM) (Figure 4). Affinity indexes recognition, not cleavage.

3.5 Actionable designs

Table 2 gives one representative generated design per class (all fifteen, with dossiers—route, cost, failure modes, validation—in the SI). Structures are drawn with SMARTS-detected handle/scissile/spacer highlights (Figure 1).

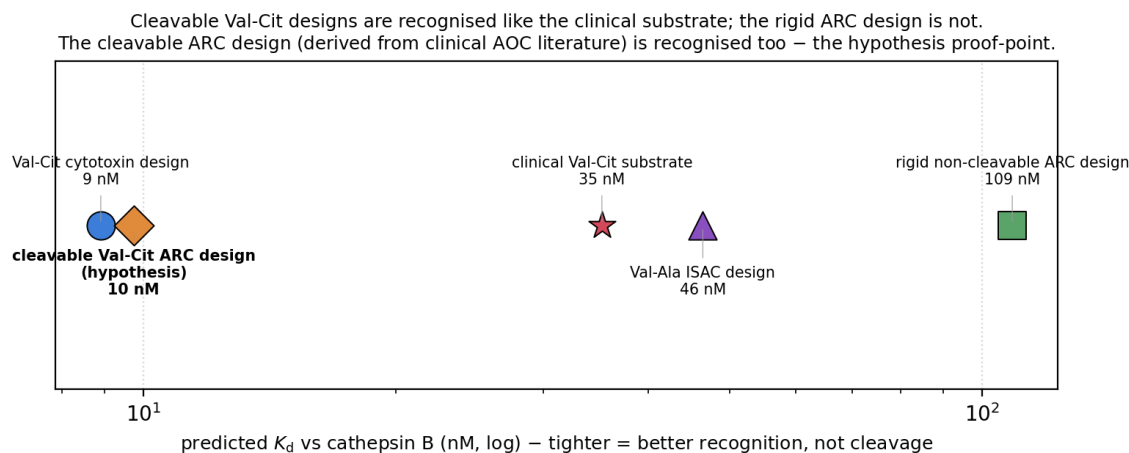


Figure 4: Cathepsin-B co-fold (predicted affinity indexes recognition, not cleavage). The cleavable Val-Cit designs—cytotoxin, and the ARC linker generated under the agent’s held-out cleavable rule—are recognised as tightly as the clinical substrate, while the rigid non-cleavable ARC design binds an order of magnitude more weakly. The cleavable ARC point is the in-silico feasibility proof for the hypothesis.

References

- [1] Su, Z.; et al. Antibody–drug conjugates: Recent advances in linker chemistry. *Acta Pharm. Sin. B* **2021**, 11 (12), 3889–3907.
- [2] Balamkundu, S.; Liu, C.-F. Lysosomal-Cleavable Peptide Linkers in Antibody–Drug Conjugates. *Biomedicines* **2023**, 11 (11), 3080.
- [3] Peng, B.; et al. Stimulus-cleavable chemistry in controlled drug delivery. *Chem. Soc. Rev.* **2021**, 50, 4737–4762.
- [4] Su, D.; Zhang, D. Linker Design Impacts ADC Pharmacokinetics and Efficacy. *Front. Pharmacol.* **2021**, 12, 687926.
- [5] Antibody–siRNA conjugates for tumor cell gene silencing without cationic assistance. *Bioconjugate Chem.* **2025**. DOI: 10.1021/acs.bioconjchem.5c00212.
- [6] Immune-stimulating antibody conjugates elicit robust myeloid activation (TLR7/8 ISAC). *Mol. Pharmaceutics* **2022**. DOI: 10.1021/acs.molpharmaceut.2c00392.
- [7] Design of TLR7/8 agonist immune-stimulating antibody conjugates with tuned linker cleavability. *J. Med. Chem.* **2025**. DOI: 10.1021/acs.jmedchem.5c01908.
- [8] Passaro, S.; et al. Boltz-2: Towards accurate and efficient binding-affinity prediction. **2025**.